

Performance Testing

Date	29 june 26
Team ID	xxxxxx
Project Name	Opticrop: Smart Agricultural production optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing and Stress Testing
Target Module / API	Crop Recommendation and Yield Prediction API
Test Environment	Local Server (Python Flask + SQLite)
Test Date	29 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User login and dashboard access	20	60	Dashboard loads within 2 seconds
2	Crop recommendation API requests	50	120	Recommendations generated successfully
3	Weather and soil data processing	40	90	Data processed without errors
4	Peak load testing for all modules	100	180	System remains stable under heavy load

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Average response time within limits
2	Response Time (Max)	< 5 seconds	3.8 seconds	Pass	Maximum response time acceptable
3	Throughput (Req/sec)		85 req/sec	Pass	High request handling capability
4	Error Rate	< 1%	0.3%	Pass	Very low error rate
5	CPU Utilization	< 80%	68%	Pass	CPU usage under threshold
6	Memory Utilization	< 80%	72%	Pass	Memory usage within limits

Step 4: Observations & Analysis

Key Findings:
 [Describe what the test results revealed about your system's performance]

Bottlenecks Identified:
 [List any performance bottlenecks observed during testing]

Optimization Steps Taken:
 [Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]

Observations & Analysis: The Opticrop system maintained stable performance during load testing.

Response times remained within acceptable limits and no significant failures were observed during peak traffic simulation.